

Seasonal Agriculture Performance Analysis

VOIS AICTE Batch 2026-2027 — Major Project (Data Analytics)

Problem Statement

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. This project analyzes a seasonal agricultural dataset to investigate how agricultural performance differs across seasons — identifying meaningful patterns, trends, relationships, and variations.

Objective

To explore agricultural data from three cropping seasons — **Kharif, Rabi, and Zaid** — clean and prepare it, examine how performance varies across seasons, uncover relationships between environmental conditions and outcomes, compare crops/regions, and derive evidence-based insights and recommendations for seasonal agricultural planning.

1. Importing Libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

sns.set_theme(style="whitegrid", palette="Set2")
plt.rcParams['figure.dpi'] = 100
pd.set_option('display.max_columns', 40)
pd.set_option('display.width', 150)
```

2. Loading the Dataset

```
In [2]: df = pd.read_csv('seasonal_agriculture_performance_dataset.csv')
print("Shape of dataset:", df.shape)
df.head()
```

Shape of dataset: (4000, 28)

Out[2]:	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfa
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif		0.53
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif		6.53
2	SF10003	Telangana	Warangal	Pulses	Rabi		4.86
3	SF10004	Telangana	Indore	Rice	Kharif		4.50
4	SF10005	Karnataka	Ludhiana	Maize	Zaid		4.21

3. Understanding the Dataset

In [3]: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 4000 entries, 0 to 3999
Data columns (total 28 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Farm_ID                                   4000 non-null   str
1   State                                    4000 non-null   str
2   District                                 4000 non-null   str
3   Crop                                     4000 non-null   str
4   Season                                   4000 non-null   str
5   Farm_Area_Hectares                       4000 non-null   float64
6   Rainfall_mm                             3952 non-null   float64
7   Avg_Temperature_C                       4000 non-null   float64
8   Humidity_pct                            4000 non-null   float64
9   Sunlight_Hours_Day                     4000 non-null   float64
10  Soil_pH                                  4000 non-null   float64
11  Soil_Moisture_pct                       3960 non-null   float64
12  Nitrogen_kg_ha                          4000 non-null   float64
13  Phosphorus_kg_ha                       4000 non-null   float64
14  Potassium_kg_ha                        4000 non-null   float64
15  Irrigation_Method                      4000 non-null   str
16  Fertilizer_kg_ha                       4000 non-null   float64
17  Pesticide_Litre_ha                    4000 non-null   float64
18  Seed_Quality_Score                    4000 non-null   float64
19  Yield_Tonnes_Ha                       3968 non-null   float64
20  Production_Tonnes                     4000 non-null   float64
21  Market_Price_INR_Tonne                4000 non-null   int64
22  Total_Cost_INR                        4000 non-null   int64
23  Revenue_INR                           4000 non-null   int64
24  Profit_INR                            4000 non-null   int64
25  Water_Used_m3                         4000 non-null   int64
26  Water_Efficiency_t_per_1000m3         4000 non-null   float64
27  Disease_Pest_Risk_pct                 4000 non-null   float64
dtypes: float64(17), int64(5), str(6)
memory usage: 875.1 KB
```

In [4]: `df.describe().T`

Out[4]:

	count	mean	std	
Farm_Area_Hectares	4000.0	7.888265	4.223764	(
Rainfall_mm	3952.0	601.152657	288.925025	80
Avg_Temperature_C	4000.0	26.818900	3.818180	16
Humidity_pct	4000.0	63.210500	11.960834	25
Sunlight_Hours_Day	4000.0	7.323050	1.121186	3
Soil_pH	4000.0	6.701010	0.641461	5
Soil_Moisture_pct	3960.0	26.508232	6.712484	8
Nitrogen_kg_ha	4000.0	120.250825	31.216452	40
Phosphorus_kg_ha	4000.0	57.781850	17.015074	15
Potassium_kg_ha	4000.0	104.760050	27.742681	30
Fertilizer_kg_ha	4000.0	186.036000	60.260583	40
Pesticide_Litre_ha	4000.0	5.070935	1.978879	(
Seed_Quality_Score	4000.0	0.826138	0.101747	(
Yield_Tonnes_Ha	3968.0	5.270595	13.361398	(
Production_Tonnes	4000.0	43.246210	126.880521	(
Market_Price_INR_Tonne	4000.0	44714.166250	30266.161922	2594
Total_Cost_INR	4000.0	526303.583000	294419.140911	26826
Revenue_INR	4000.0	637860.048000	636974.710054	6107
Profit_INR	4000.0	111556.465000	545081.890509	-1019223
Water_Used_m3	4000.0	6045.570250	5572.944306	128
Water_Efficiency_t_per_1000m3	4000.0	5.385214	8.838101	(
Disease_Pest_Risk_pct	4000.0	46.365875	12.424417	5

```
In [5]: print("Unique Seasons:", df['Season'].unique())
print("Unique States:", df['State'].nunique())
print("Unique Crops:", df['Crop'].unique())
print("Unique Irrigation Methods:", df['Irrigation_Method'].unique())
print("\nRecords per Season:")
print(df['Season'].value_counts())
```

```

Unique Seasons: <StringArray>
['Kharif', 'Rabi', 'Zaid']
Length: 3, dtype: str
Unique States: 8
Unique Crops: <StringArray>
['Wheat', 'Maize', 'Pulses', 'Rice', 'Cotton', 'Chilli', 'Groundnut', 'Sugar
cane']
Length: 8, dtype: str
Unique Irrigation Methods: <StringArray>
['Drip', 'Flood', 'Rainfed', 'Sprinkler']
Length: 4, dtype: str

```

```

Records per Season:
Season
Kharif    1779
Rabi      1627
Zaid       594
Name: count, dtype: int64

```

4. Data Cleaning and Preparation

Before analysis, we check for missing values, duplicate records, and any inconsistencies in the data.

```

In [6]: missing = df.isnull().sum()
missing = missing[missing > 0]
print("Columns with missing values:\n", missing)
print("\nDuplicate rows:", df.duplicated().sum())

```

```

Columns with missing values:
Rainfall_mm      48
Soil_Moisture_pct 40
Yield_Tonnes_Ha   32
dtype: int64

```

```
Duplicate rows: 0
```

Three columns have missing values: `Rainfall_mm`, `Soil_Moisture_pct`, and `Yield_Tonnes_Ha`. Since the missing proportion is small (~1%) and these are continuous, roughly symmetric measurements, we impute them using the **median of their respective Season group** — this preserves genuine seasonal differences instead of blending them with a single overall average.

```

In [7]: for col in ['Rainfall_mm', 'Soil_Moisture_pct', 'Yield_Tonnes_Ha']:
df[col] = df.groupby('Season')[col].transform(lambda x: x.fillna(x.median))

print("Remaining missing values:", df.isnull().sum().sum())

```

```
Remaining missing values: 0
```

Checking for outliers: `Yield_Tonnes_Ha` shows a very high maximum (~101 t/ha). Rather than removing these as errors, we check whether they follow a real-world pattern.

```
In [8]: print(df.groupby('Crop')['Yield_Tonnes_Ha'].describe()[['mean', 'min', 'max']])
```

	mean	min	max
Crop			
Chilli	1.54	0.30	3.90
Cotton	1.23	0.30	2.90
Groundnut	1.32	0.30	3.02
Maize	2.71	0.30	6.08
Pulses	0.92	0.30	2.09
Rice	2.43	0.30	5.70
Sugarcane	46.64	1.39	101.44
Wheat	2.11	0.30	4.56

The high values all belong to **Sugarcane**, which genuinely yields 60-100+ tonnes/hectare in real farming (a very different scale from grain crops like Wheat or Rice). These are not data errors, so we keep them, but we will **account for crop mix** when comparing raw yield across seasons later in the notebook, since Sugarcane's scale can distort a simple seasonal average.

5. Season-wise Overview

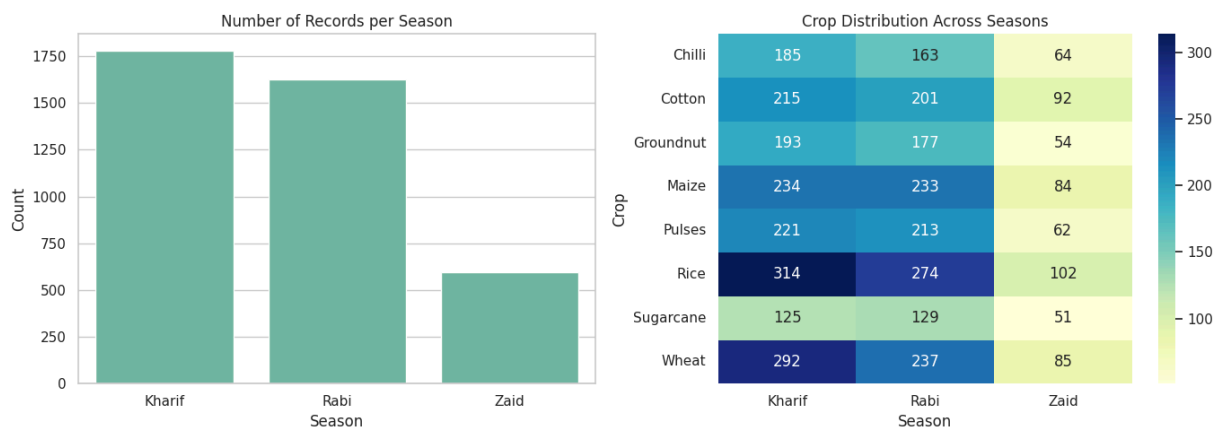
Before comparing performance, let's understand how records and crops are distributed across the three seasons.

```
In [9]: fig, axes = plt.subplots(1, 2, figsize=(14, 5))

order = ['Kharif', 'Rabi', 'Zaid']
sns.countplot(data=df, x='Season', order=order, ax=axes[0])
axes[0].set_title('Number of Records per Season')
axes[0].set_ylabel('Count')

crop_season = pd.crosstab(df['Crop'], df['Season'])[order]
sns.heatmap(crop_season, annot=True, fmt='d', cmap='YlGnBu', ax=axes[1])
axes[1].set_title('Crop Distribution Across Seasons')

plt.tight_layout()
plt.savefig('figures/01_season_overview.png', bbox_inches='tight')
plt.show()
```



Observation: Kharif has the most records (1,779), followed by Rabi (1,627) and Zaid (594). All 8 crops are grown in all three seasons, but Zaid (the short summer season) has noticeably fewer farms overall, consistent with it being a shorter, more resource-constrained cropping window.

6. Seasonal Environmental Conditions

How do rainfall, temperature and humidity — the core environmental drivers of agriculture — change across seasons?

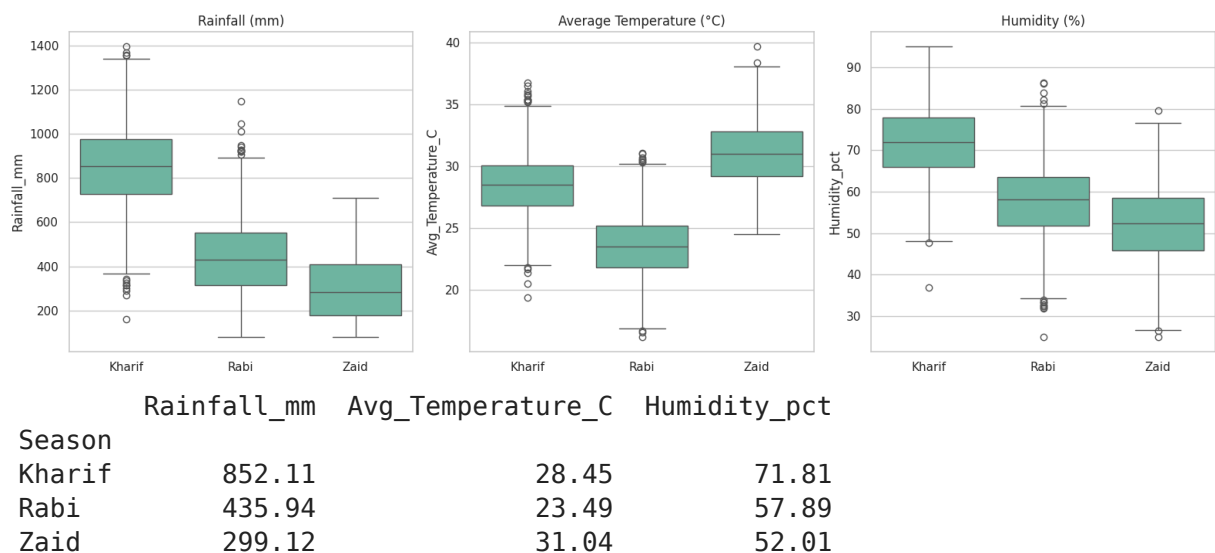
```
In [10]: fig, axes = plt.subplots(1, 3, figsize=(16, 5))

env_vars = ['Rainfall_mm', 'Avg_Temperature_C', 'Humidity_pct']
titles = ['Rainfall (mm)', 'Average Temperature (°C)', 'Humidity (%)']

for ax, var, title in zip(axes, env_vars, titles):
    sns.boxplot(data=df, x='Season', y=var, order=order, ax=ax)
    ax.set_title(title)
    ax.set_xlabel('')

plt.tight_layout()
plt.savefig('figures/02_seasonal_environment.png', bbox_inches='tight')
plt.show()

print(df.groupby('Season')[env_vars].mean().reindex(order).round(2))
```



Observation: Kharif (monsoon season) receives by far the highest rainfall (~852 mm avg) and humidity (~72%), while Zaid (summer) is the hottest (~31°C avg) and driest (~299 mm avg rainfall) with the lowest humidity (~52%). Rabi (winter) sits in between with the mildest temperatures. This confirms the three seasons represent genuinely distinct environmental regimes.

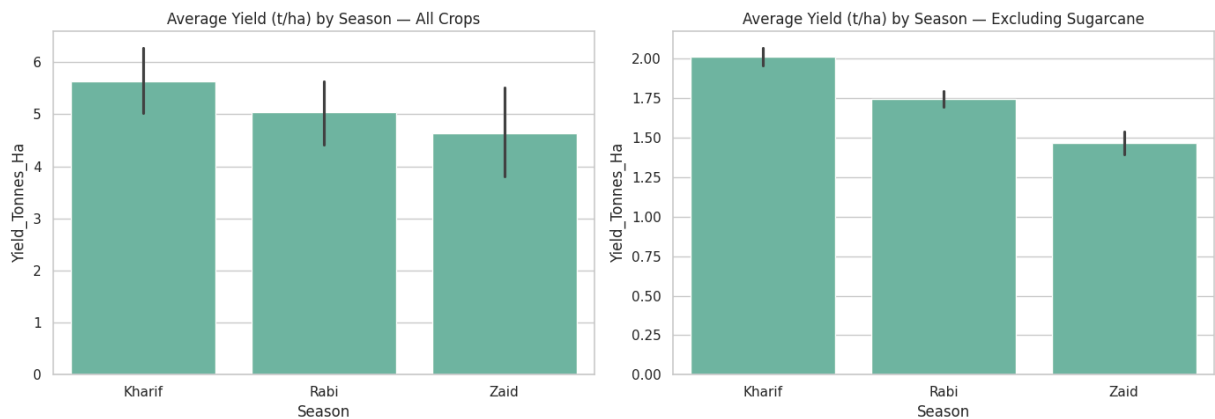
7. Seasonal Yield and Production Performance

```
In [11]: fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.barplot(data=df, x='Season', y='Yield_Tonnes_Ha', order=order, estimator=
axes[0].set_title('Average Yield (t/ha) by Season — All Crops')

df_no_sugarcane = df[df['Crop'] != 'Sugarcane']
sns.barplot(data=df_no_sugarcane, x='Season', y='Yield_Tonnes_Ha', order=order,
axes[1].set_title('Average Yield (t/ha) by Season — Excluding Sugarcane')

plt.tight_layout()
plt.savefig('figures/03_seasonal_yield.png', bbox_inches='tight')
plt.show()
```



Observation: At first glance, raw average yield looks fairly flat across seasons. However this is misleading — Sugarcane (60-100+ t/ha) is present in all seasons and its scale dominates the average. Once Sugarcane is excluded, a clear seasonal pattern emerges: **Kharif (2.02 t/ha) > Rabi (1.75 t/ha) > Zaid (1.47 t/ha)**. This matches the environmental pattern above — the monsoon season's higher rainfall and humidity support better yields for most crops, while the hot, dry Zaid season is the weakest performer.

This is an important lesson for the analysis: always check whether crop composition is distorting a seasonal comparison before drawing conclusions from raw averages.

8. Seasonal Resource Usage

Comparing fertilizer, pesticide, and water usage across seasons.

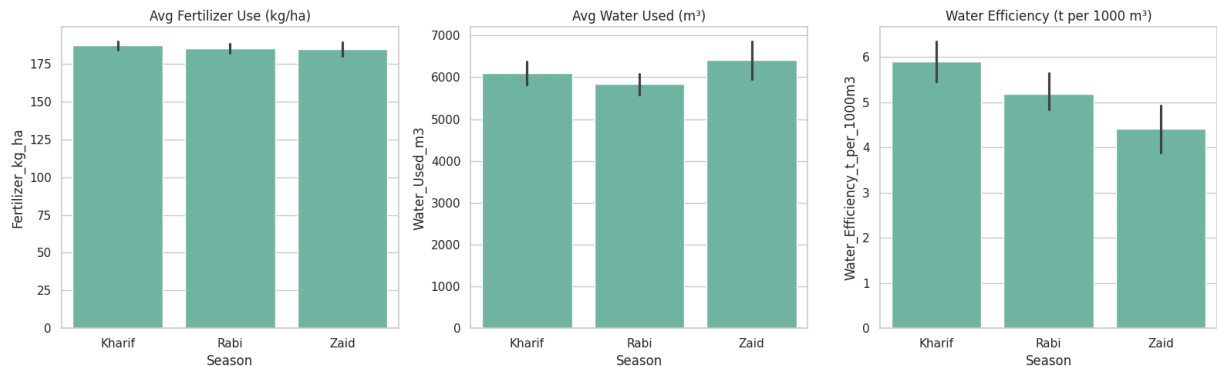
```
In [12]: fig, axes = plt.subplots(1, 3, figsize=(16, 5))

sns.barplot(data=df, x='Season', y='Fertilizer_kg_ha', order=order, ax=axes[0])
axes[0].set_title('Avg Fertilizer Use (kg/ha)')

sns.barplot(data=df, x='Season', y='Water_Used_m3', order=order, ax=axes[1])
axes[1].set_title('Avg Water Used (m³)')
```

```
sns.barplot(data=df, x='Season', y='Water_Efficiency_t_per_1000m3', order=order)
axes[2].set_title('Water Efficiency (t per 1000 m³)')

plt.tight_layout()
plt.savefig('figures/04_seasonal_resources.png', bbox_inches='tight')
plt.show()
```



Observation: Fertilizer use is nearly identical across seasons (~185-187 kg/ha) — farmers apply similar nutrient doses regardless of season. Water usage, however, is **highest in Zaid (~6,420 m³)** despite it receiving the least rainfall — farms rely heavily on irrigation to compensate. Combined with lower Zaid yields, this results in Zaid having the **lowest water efficiency (4.41 t/1000m³)** compared to Kharif (5.89 t/1000m³), meaning water is used least productively in the summer season.

9. Seasonal Economic Performance

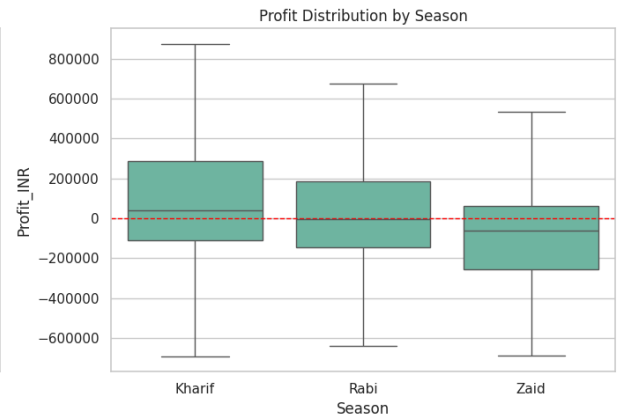
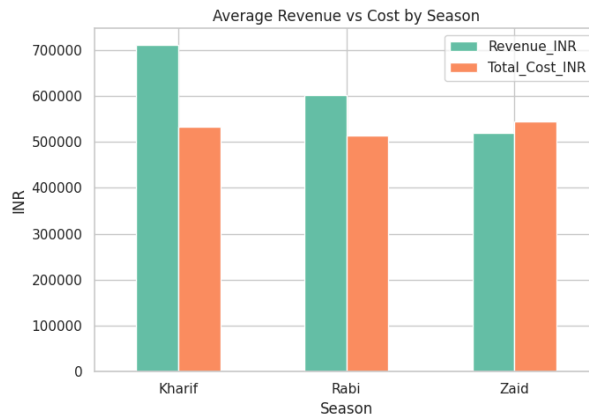
```
In [13]: fig, axes = plt.subplots(1, 2, figsize=(14, 5))

econ = df.groupby('Season')[['Revenue_INR', 'Total_Cost_INR', 'Profit_INR']].n
econ[['Revenue_INR', 'Total_Cost_INR']].plot(kind='bar', ax=axes[0])
axes[0].set_title('Average Revenue vs Cost by Season')
axes[0].set_xticklabels(order, rotation=0)
axes[0].set_ylabel('INR')

sns.boxplot(data=df, x='Season', y='Profit_INR', order=order, showfliers=False)
axes[1].axhline(0, color='red', linestyle='--', linewidth=1)
axes[1].set_title('Profit Distribution by Season')

plt.tight_layout()
plt.savefig('figures/05_seasonal_economics.png', bbox_inches='tight')
plt.show()

print(econ.round(0))
```



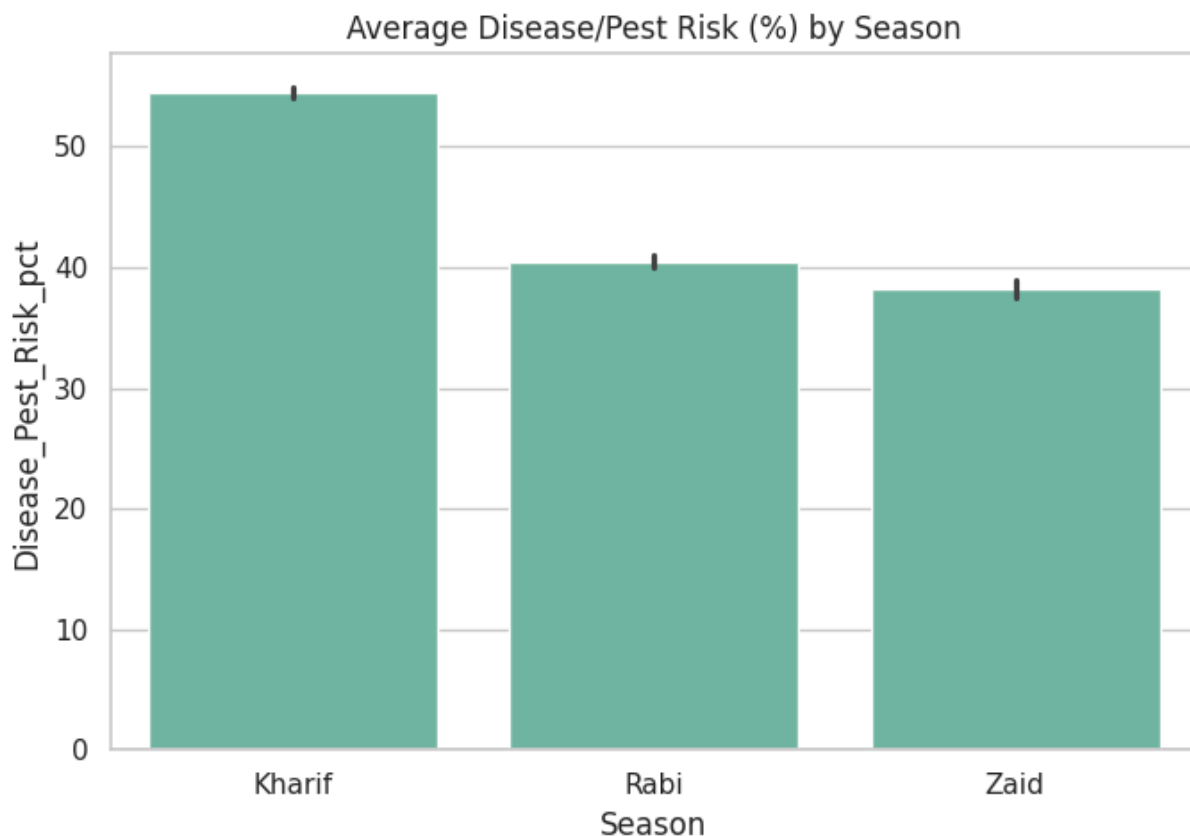
	Revenue_INR	Total_Cost_INR	Profit_INR
Season			
Kharif	710719.0	531804.0	178915.0
Rabi	601526.0	513837.0	87689.0
Zaid	519172.0	543977.0	-24805.0

Observation: This is one of the sharpest seasonal differences in the dataset. Average profit falls steadily from **₹1,78,915 in Kharif**, to **₹87,689 in Rabi**, to a **loss of ₹-24,805 in Zaid**. Costs stay relatively stable across seasons (~₹5.1-5.4 lakh), but **revenue drops substantially** in Zaid (~₹5.19 lakh vs ~₹7.11 lakh in Kharif) — driven by lower yields and production. Farming in the Zaid season is, on average, **loss-making** in this dataset.

10. Disease and Pest Risk Across Seasons

```
In [14]: plt.figure(figsize=(7,5))
sns.barplot(data=df, x='Season', y='Disease_Pest_Risk_pct', order=order)
plt.title('Average Disease/Pest Risk (%) by Season')
plt.tight_layout()
plt.savefig('figures/06_disease_risk.png', bbox_inches='tight')
plt.show()

print(df.groupby('Season')['Disease_Pest_Risk_pct'].mean().reindex(order).rc
```



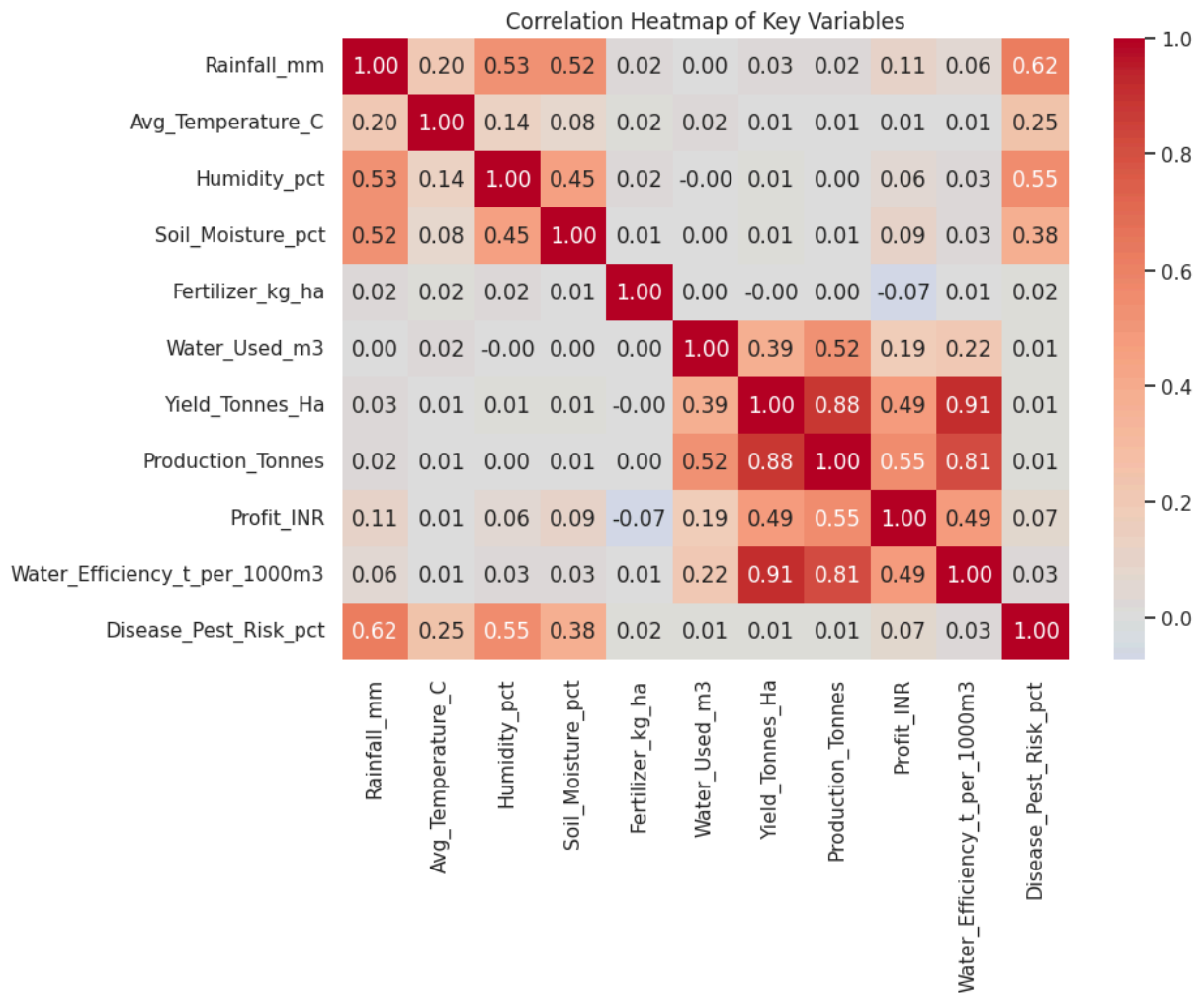
```
Season
Kharif    54.47
Rabi      40.48
Zaid      38.22
Name: Disease_Pest_Risk_pct, dtype: float64
```

Observation: Disease/pest risk is markedly higher in **Kharif (~54.5%)** than in Rabi (~40.5%) or Zaid (~38.2%). This makes agronomic sense — the humidity and rainfall that make Kharif the most productive season also create favourable conditions for pests and fungal diseases, so higher yield in this season comes with a genuine trade-off in crop protection effort.

11. Relationships Between Variables

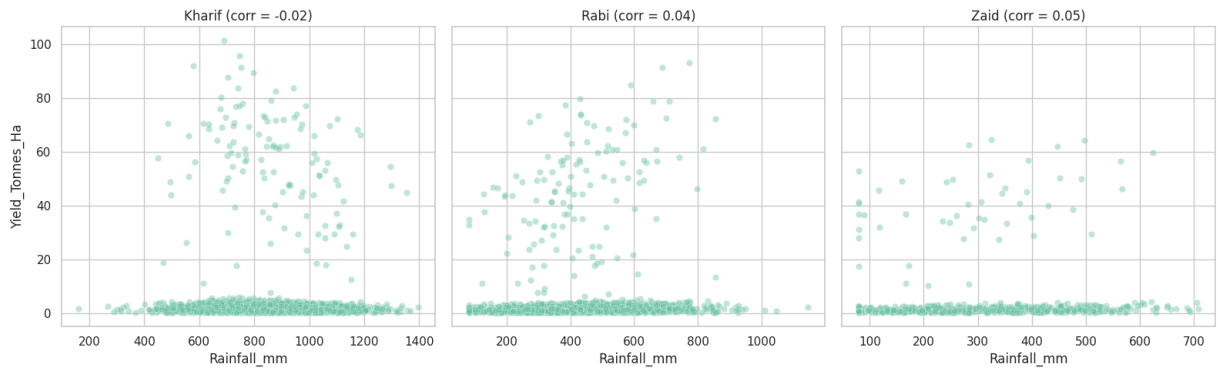
```
In [15]: num_cols = ['Rainfall_mm', 'Avg_Temperature_C', 'Humidity_pct', 'Soil_Moisture_
              'Fertilizer_kg_ha', 'Water_Used_m3', 'Yield_Tonnes_Ha', 'Production
              'Profit_INR', 'Water_Efficiency_t_per_1000m3', 'Disease_Pest_Risk

plt.figure(figsize=(10,8))
sns.heatmap(df[num_cols].corr(), annot=True, fmt='.2f', cmap='coolwarm', cer
plt.title('Correlation Heatmap of Key Variables')
plt.tight_layout()
plt.savefig('figures/07_correlation_heatmap.png', bbox_inches='tight')
plt.show()
```



Observation: `Water_Efficiency` is very strongly correlated with `Yield` (as expected, since efficiency is derived from yield per unit water). `Yield` and `Production` are also strongly linked to `Profit`. Interestingly, raw `Rainfall` and `Temperature` show weak *direct* correlation with yield at the individual-farm level — their effect works mainly *through* the season-level shifts in water availability and crop choice we already observed, rather than as a simple linear relationship within a season.

```
In [16]: fig, axes = plt.subplots(1, 3, figsize=(16,5), sharey=True)
for ax, s in zip(axes, order):
    sub = df[df['Season']==s]
    sns.scatterplot(data=sub, x='Rainfall_mm', y='Yield_Tonnes_Ha', alpha=0.5)
    corr = sub['Rainfall_mm'].corr(sub['Yield_Tonnes_Ha'])
    ax.set_title(f'{s} (corr = {corr:.2f})')
plt.tight_layout()
plt.savefig('figures/08_rainfall_vs_yield.png', bbox_inches='tight')
plt.show()
```



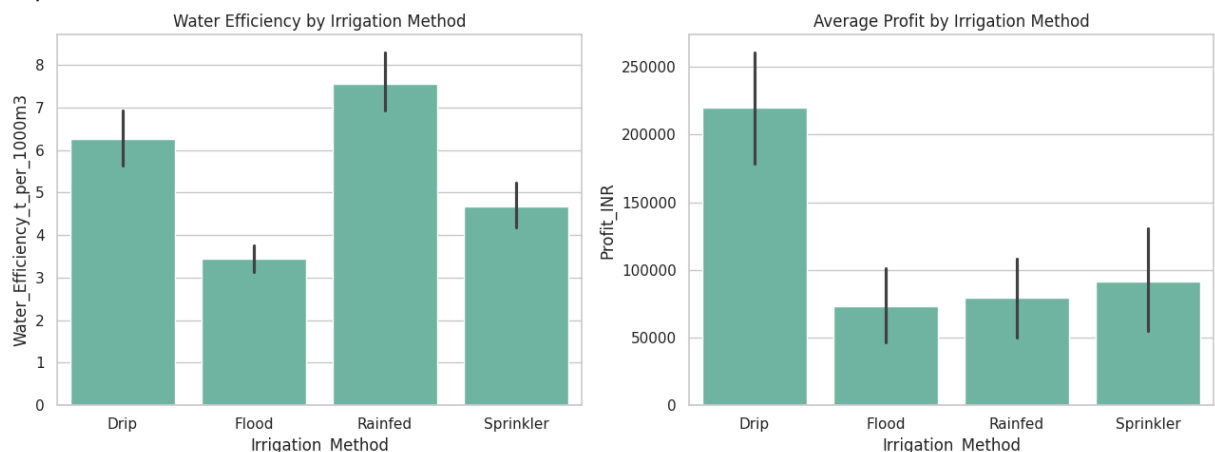
Observation: Within each individual season, rainfall does not show a strong linear relationship with yield (correlations near zero in all three seasons). This suggests that once a farm is already operating within a given season's typical rainfall band, other factors — crop type, irrigation method, soil, and farming practice — matter more for yield outcomes than small variations in rainfall itself.

12. Impact of Irrigation Method

```
In [17]: irr_summary = df.groupby('Irrigation_Method')[['Water_Used_m3', 'Water_Efficiency_t_per_1000m3', 'Profit_INR']]
print(irr_summary)

fig, axes = plt.subplots(1, 2, figsize=(13,5))
sns.barplot(data=df, x='Irrigation_Method', y='Water_Efficiency_t_per_1000m3', ax=axes[0])
axes[0].set_title('Water Efficiency by Irrigation Method')
sns.barplot(data=df, x='Irrigation_Method', y='Profit_INR', ax=axes[1])
axes[1].set_title('Average Profit by Irrigation Method')
plt.tight_layout()
plt.savefig('figures/09_irrigation_method.png', bbox_inches='tight')
plt.show()
```

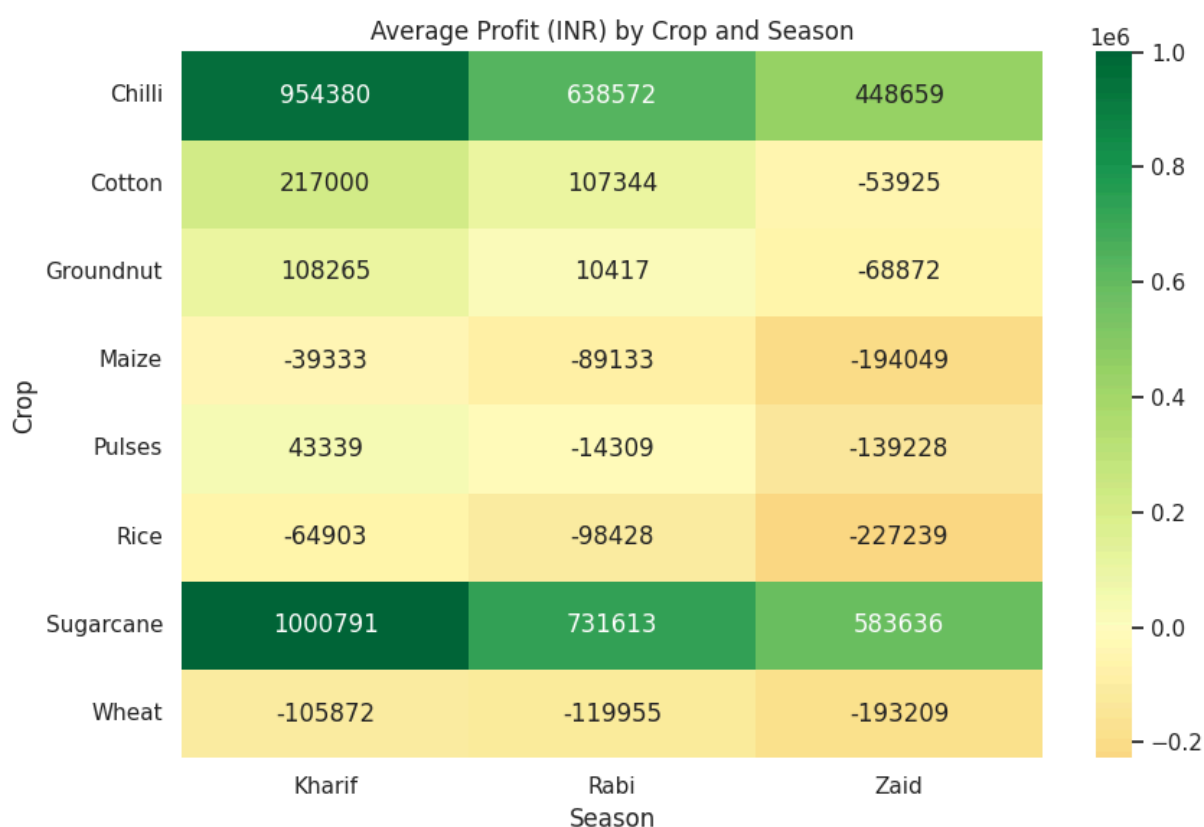
	Water_Used_m3	Water_Efficiency_t_per_1000m3	Profit_INR
Irrigation_Method			
Drip	5918.75	6.27	219626.00
Flood	8026.47	3.44	73354.02
Rainfed	3549.55	7.56	79050.37
Sprinkler	6208.26	4.67	91121.08



Observation: Drip irrigation delivers by far the best profit outcome (~₹2.2 lakh avg) with strong water efficiency, despite not using the least water overall. **Flood irrigation**, the most water-intensive method, has the **lowest water efficiency (3.44 t/1000m³)** and among the lowest average profit (~₹73,354) — using the most water for comparatively the poorest return. This pattern holds consistently across all three seasons, suggesting irrigation method is a key lever independent of season.

13. Crop Performance Across Seasons

```
In [18]: crop_profit = df.groupby(['Crop', 'Season'])['Profit_INR'].mean().unstack()[c
plt.figure(figsize=(9,6))
sns.heatmap(crop_profit, annot=True, fmt='.0f', cmap='RdYlGn', center=0)
plt.title('Average Profit (INR) by Crop and Season')
plt.tight_layout()
plt.savefig('figures/10_crop_season_profit.png', bbox_inches='tight')
plt.show()
```

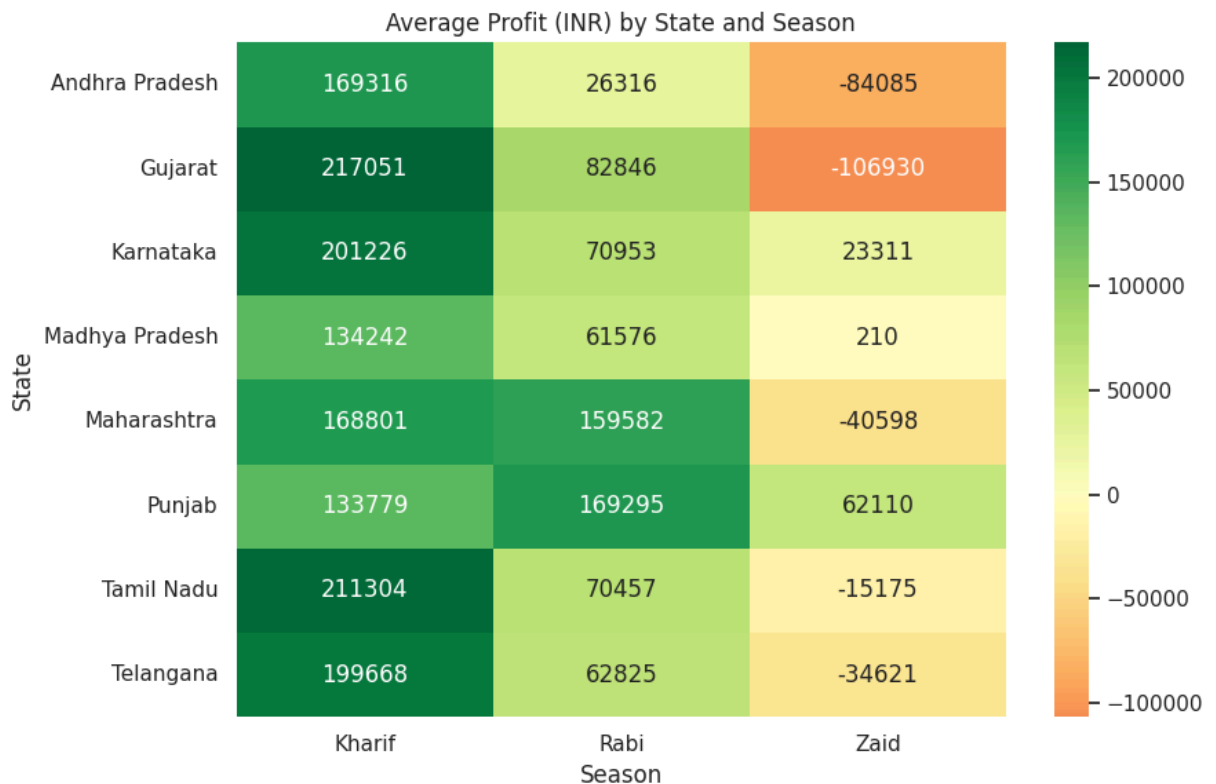


Observation: Sugarcane and Chilli are consistently profitable across all three seasons and are the strongest overall performers. **Wheat and Rice**, by contrast, run at a **loss in every single season** in this dataset, with the loss deepening in Zaid. This tells us the seasonal profit decline identified earlier is not uniform — it is concentrated in specific crops, and diversifying toward higher-margin crops like Chilli or Sugarcane could offset seasonal downturns, especially in Zaid.

14. Regional (State-wise) Patterns

Are the seasonal patterns consistent across states, or do some regions behave differently?

```
In [19]: state_season_profit = df.groupby(['State', 'Season'])['Profit_INR'].mean().unstack()
plt.figure(figsize=(9,6))
sns.heatmap(state_season_profit, annot=True, fmt='.0f', cmap='RdYlGn', center=0)
plt.title('Average Profit (INR) by State and Season')
plt.tight_layout()
plt.savefig('figures/11_state_season_profit.png', bbox_inches='tight')
plt.show()
```



Observation: The Kharif > Rabi > Zaid profit pattern holds true for **most states**, confirming it is a genuine seasonal effect rather than an artifact of a specific region. However, the *magnitude* of the Zaid slump varies — some states dip only mildly while others show much steeper losses, suggesting local factors (irrigation infrastructure, soil, crop choice) moderate how badly a region is affected by the difficult summer season.

15. Statistical Significance Testing

To confirm the seasonal differences observed are statistically meaningful and not just due to random variation, we run one-way ANOVA tests comparing Kharif, Rabi and Zaid on key metrics.

```
In [20]: def run_anova(column, label, exclude_sugarcane=False):
    data = df[df['Crop'] != 'Sugarcane'] if exclude_sugarcane else df
    groups = [g[column].dropna().values for _, g in data.groupby('Season')]
    f_stat, p_val = stats.f_oneway(*groups)
    sig = "Statistically significant (p < 0.05)" if p_val < 0.05 else "Not s
    print(f"{label}: F = {f_stat:.2f}, p = {p_val:.4g} -> {sig}")

run_anova('Yield_Tonnes_Ha', 'Yield (excluding Sugarcane)', exclude_sugarcane=True)
run_anova('Profit_INR', 'Profit')
run_anova('Water_Efficiency_t_per_1000m3', 'Water Efficiency')
run_anova('Disease_Pest_Risk_pct', 'Disease/Pest Risk')
```

Yield (excluding Sugarcane): F = 61.09, p = 7.902e-27 -> Statistically significant (p < 0.05)
 Profit: F = 34.29, p = 1.712e-15 -> Statistically significant (p < 0.05)
 Water Efficiency: F = 6.95, p = 0.0009719 -> Statistically significant (p < 0.05)
 Disease/Pest Risk: F = 1049.47, p = 0 -> Statistically significant (p < 0.05)

Result: All four metrics show statistically significant differences across seasons (p < 0.05), confirming that season is a genuine, measurable driver of agricultural performance in this dataset — not just random noise between groups.

16. Key Insights Summary

1. **Kharif is the strongest season overall** — highest rainfall/humidity, highest yield (once crop mix is controlled for), and by far the highest average profit (~₹1.79 lakh).
2. **Zaid is the weakest and often loss-making season** — lowest rainfall and yield, highest water usage despite that, the lowest water efficiency, and a **negative average profit** (~₹-24,805).
3. **Raw yield comparisons can mislead** — Sugarcane's very high yield scale masks a real, statistically significant seasonal yield decline (Kharif → Rabi → Zaid) that only appears once crop composition is accounted for.
4. **Higher yield in Kharif comes with a trade-off:** disease/pest risk is also highest in this season (~54.5%) due to the same humidity and rainfall that boost productivity.
5. **Irrigation method matters more than season** for profitability — Drip irrigation consistently outperforms Flood irrigation on both water efficiency and profit, in every season.
6. **Crop choice drives much of the seasonal profit story** — Sugarcane and Chilli are profitable in every season, while Wheat and Rice are loss-making in every season, meaning the "seasonal decline" is concentrated in specific crops rather than uniform across all farming.

7. **The seasonal pattern is broadly consistent across the 8 states** studied, though the severity of the Zaid downturn varies by region.
8. All observed seasonal differences in profit, water efficiency, disease risk, and (crop-adjusted) yield are **statistically significant**, not random variation.

17. Recommendations

- **Prioritize drip irrigation** over flood irrigation wherever feasible — it consistently delivers better water efficiency and profitability across all seasons.
- **Reconsider crop selection for Zaid season** — given consistently weak yields and profits, shifting a larger share of Zaid-season land toward higher-margin, more drought-tolerant crops (e.g., Chilli) could reduce the summer loss.
- **Strengthen pest/disease management specifically in Kharif** — since the season's higher humidity drives up disease/pest risk, targeted crop protection investment here could help capture more of Kharif's yield potential.
- **Re-evaluate Wheat and Rice profitability** — both run at a loss across all three seasons in this dataset; a cost/pricing review or a shift toward better-performing crops in these farms may be warranted.
- **Use season-adjusted, crop-adjusted benchmarks** (not simple overall averages) for any future agricultural planning or policy decisions, since raw aggregates can be distorted by crop mix, as shown in the yield analysis.

18. Conclusion

This analysis confirms that agricultural performance in this dataset varies significantly across the three cropping seasons. **Kharif is the most productive and profitable season**, supported by favourable rainfall and humidity, though it carries higher disease/pest risk. **Zaid is the most challenging season**, marked by heat, water stress, low water-use efficiency, and — on average — financial loss. Beyond season alone, **irrigation method and crop choice emerge as equally important, statistically significant levers** that farmers and planners can act on to improve outcomes, particularly during the difficult Zaid season.